

R Programming Tutorial

#05

Hypothesis Testing

#lectureFIVE-TWO

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#05-1

HYPOTHESIS TESTING

DECISION CRITERION

Hypothesis Testing

Decision Criterion

Before we continue, let's recall what is decision criterion.
 Please refer to #lectureFIVE (slide 26-37)



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Decision Criterion

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- Traditional method:
 - If the test statistic falls within the critical region, **reject H_0** .
 - If the test statistic does not fall within the critical region, **fail to reject H_0** .

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Decision Criterion

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- P-value method:

Using the significance level α :

 - If $P\text{-value} \leq \alpha$, **reject H_0** .
 - If $P\text{-value} > \alpha$, **fail to reject H_0** .

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Also called CRITICAL REGION METHOD

Hypothesis Testing

Decision Criterion

METHOD 1: TRADITIONAL

1. Identify critical region
2. Calculate test statistics
3. If test statistics fall within critical region, reject H_0
4. If test statistics does not fall within critical region, fail to reject H_0

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Decision Criterion

- Traditional method:
 - If the test statistic falls within the critical region, **reject H_0** .
 - If the test statistic does not fall within the critical region, **fail to reject H_0** .

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METHOD 2: P-VALUE

1. Define significance level.
2. Calculate test statistics.
3. Find p-value
4. If p-value < significance level, reject H_0
5. If p-value > significance level, fail to reject H_0

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Decision Criterion

- P-value method:
Using the significance level α .
 - If $P\text{-value} \leq \alpha$, **reject H_0** .
 - If $P\text{-value} > \alpha$, **fail to reject H_0** .

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#05-2

HYPOTHESIS TESTING 1 SAMPLE IN R

PROPORTION: ONE-TAILED

Hypothesis Testing 1 Sample

Proportion: One-Tailed

PROBLEM

Suppose 60% of citizens voted in last election. 85 out of 148 people in a telephone survey said that they voted in current election. At 0.5 significance level, can we reject the null hypothesis that the proportion of voters in the population is below 60% this year?

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

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Hypothesis Testing 1 Sample

Proportion: One-Tailed (TRADITIONAL METHOD)

WHAT WE KNOW?

- Sample size, n: 148
- People who vote : 85
- $H_0: p = 0.6$ $H_1: p < 0.6$
- Significance level: 0.05

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

```
1  n = 148
2  k = 85
3  p = 0.6
4  pbar = k/n
5
6  // Calculate Z statistics
7  z = (pbar-p)/sqrt(p*(1-p)/n)
8
9  // Calculate critical value
10 alpha = 0.05
11 z.alpha = qnorm(1-alpha)-z.alpha
12
13
14
15 > z
16 [1] -0.6375983
17 > -z.alpha
18 [1] -1.644854
19
20
```

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: One-Tailed (P-VALUE METHOD)

WHAT WE KNOW?

- Sample size, n : 148
- People who vote : 85
- $H_0: p = 0.6$ $H_1: p < 0.6$
- Significance level: 0.05

pnorm() can be used to get p-value for a calculated test statistics.

Ref: <http://seankross.com/notes/dpqr/>

```
1  n = 148
2  k = 85
3  p = 0.6
4  pbar = k/n
5
6  // Calculate Z statistics
7  z = (pbar-p)/sqrt(p*(1-p)/n)
8
9  // Calculate p-value for Z
10 pval = pnorm(z)
11
12
13
14
15
16
17
18
19
20
```

[1] 0.2618676

#05-3

HYPOTHESIS TESTING 1 SAMPLE IN R

TEST ON MEAN (VARIANCE KNOWN): ONE-TAILED

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: One-Tailed

REFER to:

lectureFIVE-TWO #02

EXAMPLE (slide 51)

Example

- **Writing a Hit Song** In the manual “How to Have a Number One the Easy Way,” by KLF Publications, it is stated that a song “must be no longer than three minutes and thirty seconds” (or 210 seconds). A simple random sample of 40 current hit songs results in a mean length of 252.5 sec. (The songs are by Timberlake, Furtado, Daughtry, Stefani, Fergie, Akon, Ludacris, etc.) Assume that the standard deviation of song lengths is 54.5 sec. Use a 0.05 significance level to test the claim that the sample is from a population of songs with a mean greater than 210 sec. What do these results suggest about the advice given in the manual?

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: One-Tailed

WHAT WE KNOW?

- Sample size, n : 40
- Standard deviation: 54.5
- Sample mean: 252.5
- $H_0: \mu = 210$ $H_1: \mu > 210$
- Significance level: 0.05

$$Z = \frac{\bar{X} - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

Example

- **Writing a Hit Song** In the manual “How to Have a Number One the Easy Way,” by KLF Publications, it is stated that a song “must be no longer than three minutes and thirty seconds” (or 210 seconds). A simple random sample of 40 current hit songs results in a mean length of 252.5 sec. (The songs are by Timberlake, Furtado, Daughtry, Stefani, Fergie, Akon, Ludacris, etc.) Assume that the standard deviation of song lengths is 54.5 sec. Use a 0.05 significance level to test the claim that the sample is from a population of songs with a mean greater than 210 sec. What do these results suggest about the advice given in the manual?

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: One-Tailed (TRADITIONAL METHOD)

WHAT WE KNOW?

- Sample size, n : 40
- Standard deviation: 54.5
- Sample mean: 252.5
- $H_0: \mu = 210$ $H_1: \mu > 210$
- Significance level: 0.05

$$Z = \frac{X - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```
1 n = 40
2 sd = 54.5
3 xbar = 252.5
4 mu = 210
5 alpha = 0.05
6
7 // Calculate z test statistics
8 z = (xbar-mu)/(sd/sqrt(n))
9
10 // Calculate critical value
11 z.alpha = qnorm(1-alpha)
12
13
14
15
16
17
18
19
20
```

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: One-Tailed (TRADITIONAL METHOD)

WHAT WE KNOW?

- Sample size, n : 40
- Standard deviation: 54.5
- Sample mean: 252.5
- $H_0: \mu = 210$ $H_1: \mu > 210$
- Significance level: 0.05

$$z = \frac{\bar{X} - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```
1 n = 40
2 sd = 54.5
3 xbar = 252.5
4 mu = 210
5 alpha = 0.05
6
7 // Calculate z test statistics
8 z = (xbar-mu)/(sd/sqrt(n))
9
10 // Calculate critical value
11 z.alpha = qnorm(1-alpha)
12
13
14
15
16 > z
17 [1] 4.931993
18 > z.alpha
19 [1] 1.644854
20
```

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: One-Tailed (P-VALUE METHOD)

WHAT WE KNOW?

- Sample size, n : 40
- Standard deviation: 54.5
- Sample mean: 252.5
- $H_0: \mu = 210$ $H_1: \mu > 210$
- Significance level: 0.05

$$Z = \frac{\bar{X} - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```

1  n = 40
2  sd = 54.5
3  xbar = 252.5
4  mu = 210
5  alpha = 0.05
6
7  // Calculate z test statistics
8  z = (xbar-mu)/(sd/sqrt(n))
9
10 // Calculate p-value of z
11 pval = pnorm(z, lower.tail=FALSE)
12
13
14
  
```

pnorm() can be used to get p-value for a calculated test statistics.

Ref: <http://seankross.com/notes/dpqr/>

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```
[1] 4.069748e-07
```

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: One-Tailed (P-VALUE METHOD)

Solution

original claim: $\mu > 210$ sec

$H_0: \mu = 210$ sec

$H_1: \mu > 210$ sec

$\alpha = 0.05$

C.V. $z = z_{\alpha} = z_{0.05} = 1.645$

calculations:

$$z_{\bar{x}} = (\bar{x} - \mu_{\bar{x}}) / \sigma_{\bar{x}}$$

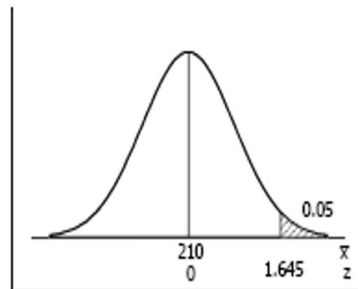
$$= (252.5 - 210) / (54.5 / \sqrt{40})$$

$$= 42.5 / 8.6172 = 4.93$$

$$\text{P-value} = P(z > 4.93) = 0.5 - 0.4999 = 0.0001$$

conclusion:

Reject H_0 ; there is sufficient evidence to conclude that $\mu > 210$. There is sufficient evidence to support the claim that the sample is from a population of songs with a mean greater than 210 seconds. These results suggest that the advice given in the manual is not good advice.



(#lectureFIVE #02, slide 52)

```

1  n = 40
2  sd = 54.5
3  xbar = 252.5
4  mu = 210
5  alpha = 0.05
6
7  // Calculate z test statistics
8  z = (xbar-mu)/(sd/sqrt(n))
9
10 // Calculate p-value of z
11 pval = pnorm(z, lower.tail=FALSE)
12
13
14
15
16
17
18 [1] 4.069748e-07
19
20
  
```

#05-4

HYPOTHESIS TESTING 1 SAMPLE IN R

TEST ON MEAN (VARIANCE KNOWN): TWO-TAILED

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: Two-Tailed

Problem:

Suppose the mean weight of King Penguins found in an Antarctic colony last year was 15.4 kg. In a sample of 35 penguins same time this year in the same colony, the mean penguin weight is 14.6 kg. Assume the population standard deviation is 2.5 kg. At .05 significance level, can we reject the null hypothesis that the mean penguin weight does not differ from last year?

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22

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: Two-Tailed (TRADITIONAL METHOD)

WHAT WE KNOW?

- Sample size, n : 35
- Standard deviation: 2.5
- Sample mean: 14.6
- $H_0: \mu = 15.4$ $H_1: \mu \neq 15.4$
- Significance level: 0.05

$$Z = \frac{X - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```
1  n = 35
2  sd = 2.5
3  xbar = 14.6
4  mu = 15.4
5  alpha = 0.05
6
7  // Calculate Z statistics
8  z = (xbar-mu)/(sd/sqrt(n))
9
10 // Calculate critical value
11 z.alpha = qnorm(1-(alpha/2))
12
13 // Display the c.v. on both sides
14 c(-z.alpha, z.alpha)
15
16
17 > Z
18 [1] -1.893146
19 > c(-z.alpha, z.alpha)
20 [1] -1.959964  1.959964
```

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: Two-Tailed (TRADITIONAL METHOD)

WHAT WE KNOW?

- Sample size, n : 35
- Standard deviation: 2.5
- Sample mean: 14.6
- $H_0: \mu = 15.4$ $H_1: \mu \neq 15.4$
- Significance level: 0.05

```
1  n = 35
2  sd = 2.5
3  xbar = 14.6
4  mu = 15.4
5  alpha = 0.05
6
7  // Calculate Z statistics
8  z = (xbar-mu)/(sd/sqrt(n))
9
10 // Calculate critical value
11 z.alpha = qnorm(1-(alpha/2))
12
13 // Display the c.v. on both sides
14 c(-z.alpha, z.alpha)
15
16
17 > z
18 [1] -1.893146
19
20 > c(-z.alpha, z.alpha)
[1] -1.959964  1.959964
```

In two tailed test, the critical value is calculated based on half of the alpha value

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: Two-Tailed (P-VALUE METHOD)

WHAT WE KNOW?

- Sample size, n: 35
- Standard deviation: 2.5
- Sample mean: 14.6
- $H_0: \mu = 15.4$ $H_1: \mu \neq 15.4$
- Significance level: 0.05

$$Z = \frac{\bar{X} - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```
1  n = 35
2  sd = 2.5
3  xbar = 14.6
4  mu = 15.4
5  alpha = 0.05
6
7  // Calculate Z statistics
8  z = (xbar-mu)/(sd/sqrt(n))
9
10 // Calculate p-value for Z
11 pval = 2*pnorm(z)
12
13
14
15 [1] 0.05833852
16
17
18
19
20
```

Hypothesis Testing 1 Sample

Population MEAN with Known Variance: Two-Tailed (P-VALUE METHOD)

WHAT WE KNOW?

- Sample size, n : 35
- Standard deviation: 2.5
- Sample mean: 14.6
- $H_0: \mu = 15.4$ $H_1: \mu \neq 15.4$
- Significance level: 0.05

P-value calculated by `pnorm()` represent one-tailed P-value. For two tailed test, the P-value must multiply by 2

```
1  n = 35
2  sd = 2.5
3  xbar = 14.6
4  mu = 15.4
5  alpha = 0.05
6
7  // Calculate Z statistics
8  z = (xbar-mu)/(sd/sqrt(n))
9
10 // Calculate p-value for Z
11 pval = 2*pnorm(z)
12
13
14
15
16
17
18 [1] 0.05833852
19
20
```

#05-5

HYPOTHESIS TESTING 1 SAMPLE IN R

TEST ON MEAN (VARIANCE UNKNOWN): ONE-TAILED

Hypothesis Testing 1 Sample

Population MEAN with Unknown Variance: One-Tailed

REFER to:

lectureFIVE-TWO #02

EXAMPLE (slide 57)

Example

The increased availability of light materials with high strength has revolutionized the design and manufacture of golf clubs, particularly drivers. Clubs with hollow heads and very thin faces can result in much longer tee shots, especially for players of modest skills. This is due partly to the “spring-like effect” that the thin face imparts to the ball. Firing a golf ball at the head of the club and measuring the ratio of the outgoing velocity of the ball to the incoming velocity can quantify this spring-like effect. The ratio of velocities is called the coefficient of restitution of the club. An experiment was performed in which **15 drivers** produced by a particular club maker were selected at random and their coefficients of restitution measured. In the experiment the golf balls were fired from an air cannon so that the incoming velocity and spin rate of the ball could be precisely controlled. **It is of interest to determine if there is evidence (with $\alpha = 0.05$) to support a claim that the mean coefficient of restitution exceeds 0.82.** The observations follow:

0.8411	0.8191	0.8182	0.8125	0.8750
0.8580	0.8532	0.8483	0.8276	0.7983
0.8042	0.8730	0.8282	0.8359	0.8660

Hypothesis Testing 1 Sample

Population MEAN with Unknown Variance: One-Tailed (TRADITIONAL METHOD)

WHAT WE KNOW?

- Sample size, n : 15
- List of data points given
- $H_0: \mu = 0.82$ $H_1: \mu > 0.82$
- Significance level: 0.05

$$T_0 = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

Example

The increased availability of light materials with high strength has revolutionized the design and manufacture of golf clubs, particularly drivers. Clubs with hollow heads and very thin faces can result in much longer tee shots, especially for players of modest skills. This is due partly to the “spring-like effect” that the thin face imparts to the ball. Firing a golf ball at the head of the club and measuring the ratio of the outgoing velocity of the ball to the incoming velocity can quantify this spring-like effect. The ratio of velocities is called the coefficient of restitution of the club. An experiment was performed in which 15 drivers produced by a particular club maker were selected at random and their coefficients of restitution measured. In the experiment the golf balls were fired from an air cannon so that the incoming velocity and spin rate of the ball could be precisely controlled. It is of interest to determine if there is evidence (with $\alpha = 0.05$) to support a claim that the mean coefficient of restitution exceeds 0.82. The observations follow:

0.8411	0.8191	0.8182	0.8125	0.8750
0.8580	0.8532	0.8483	0.8276	0.7983
0.8042	0.8730	0.8282	0.8359	0.8660

Hypothesis Testing 1 Sample

Population MEAN with Unknown Variance: One-Tailed (TRADITIONAL METHOD)

WHAT WE KNOW?

- Sample size, n: 15
- List of data points given
- $H_0: \mu = 0.82$ $H_1: \mu > 0.82$
- Significance level: 0.05

$$T_0 = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

```

1  x = c(0.8411, 0.8191, 0.8182,
2         0.8125, 0.8750, 0.8580,
3         0.8532, 0.8483, 0.8276,
4         0.7983, 0.8042, 0.8730,
5         0.8282, 0.8359, 0.8660)

```

```

6  n = 15

```

```

7  s = sd(x)

```

```

8  xbar = mean(x)

```

```

9  mu = 0.82

```

```

10

```

```

11 // Calculate t statistics

```

```

12 t = (xbar-mu)/(s/sqrt(n))

```

```

13

```

```

14 // Calculate critical value

```

```

15 alpha = 0.05

```

```

16 t.alpha = qt(1-alpha, df=n-1)

```

```

17

```

```

18 > t

```

```

19 [1] 2.718979

```

```

20

```

```

21 > t.alpha

```

```

22

```

```

[1] 1.76131

```

Hypothesis Testing 1 Sample

Population MEAN with Unknown Variance: One-Tailed (TRADITIONAL METHOD)

WHAT WE KNOW?

- Sample size, n : 15
- List of data points given
- $H_0: \mu = 0.82$ $H_1: \mu > 0.82$
- Significance level: 0.05

qt() can be used to get the T critical value.

2 parameters: 1) t statistics, 2) df

Ref: <http://stat.ethz.ch/R-manual/R-devel/library/stats/html/TDist.html>

```
1  x = c(0.8411, 0.8191, 0.8182,
2        0.8125, 0.8750, 0.8580,
3        0.8532, 0.8483, 0.8276,
4        0.7983, 0.8042, 0.8730,
5        0.8282, 0.8359, 0.8660)
6  n = 15
7  s = sd(x)
8  xbar = mean(x)
9  mu = 0.82
10
11 // Calculate t statistics
12 t = (xbar-mu)/(s/sqrt(n))
13
14 // Calculate critical value
15 alpha = 0.05
16 t.alpha = qt(1-alpha, df=n-1)
17
18
19
20
21
22
```

```
> t
[1] 2.718979
> t.alpha
[1] 1.76131
```

Hypothesis Testing 1 Sample

Population MEAN with Unknown Variance: One-Tailed (TRADITIONAL METHOD)

Solution

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- Reject H_0 if $t_0 > t_{0.05,14} = 1.761$.
- Computations: Since $\bar{x} = 0.83725$, $s = 0.02456$, $\mu_0 = 0.82$ and $n = 15$, we have

$$t_0 = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{0.83725 - 0.82}{0.02456 / \sqrt{15}} = 2.72.$$

- Conclusion: Since $t_0 = 2.72 > 1.761$, we reject H_0 : $\mu = 0.82$ at the 0.05 level of significance that the mean coefficient of restitution exceeds 0.82.

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(#lectureFIVE #02, slide 58)

```
1 x = c(0.8411, 0.8191, 0.8182,
2       0.8125, 0.8750, 0.8580,
3       0.8532, 0.8483, 0.8276,
4       0.7983, 0.8042, 0.8730,
5       0.8282, 0.8359, 0.8660)
```

```
6 n = 15
7 s = sd(x)
8 xbar = mean(x)
9 mu = 0.82
10
11 // Calculate t statistics
12 t = (xbar-mu)/(s/sqrt(n))
13
14 // Calculate critical value
15 alpha = 0.05
16 t.alpha = qt(1-alpha, df=n-1)
```

```
17
```

```
18 > t
19 [1] 2.718979
20
21 > t.alpha
22 [1] 1.76131
```

Hypothesis Testing 1 Sample

Population MEAN with Unknown Variance: One-Tailed (P-VALUE METHOD)

Population mean with Unknown Variance – One Tailed (P-value Method)

WHAT WE KNOW?

- Sample size, n : 15
- List of data points given
- $H_0: \mu = 0.82$ $H_1: \mu > 0.82$
- Significance level: 0.05

pt() is used to get the p-value of T statistics.

2-3 parameters: 1) t statistics, 2) df 3)
lower.tail settings

Ref: <http://stat.ethz.ch/R-manual/R-devel/library/stats/html/TDist.html>

```

1  x = c(0.8411, 0.8191, 0.8182, 0.8125,
2        0.8750, 0.8580, 0.8532, 0.8483,
3        0.8276, 0.7983, 0.8042, 0.8730,
4        0.8282, 0.8359, 0.8660)
5  n = 15
6  s = sd(x)
7  xbar = mean(x)
8  mu = 0.82
9
10 // Calculate t statistics
11 t = (xbar-mu)/(s/sqrt(n))
12
13 // Calculate p-value of t
14 pval = pt(t, df=n-1, lower.tail=FALSE)
15
16 [1] 0.008313369
17
18
19
20
```

#05-6

HYPOTHESIS TESTING 1 SAMPLE IN R

TEST ON VARIANCE: TWO-TAILED

Hypothesis Testing 1 Sample

Population Variance: Two-Tailed

WHAT WE KNOW?

- Sample size, n : 9
- Sample standard deviation: 3
- $H_0: \sigma = 3.7$ $H_1: \sigma < 3.7$
- Significance level: 0.05

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

```
1  n = 9
2  s.sd = 3
3  sigma = 3.7
4  alpha = 0.05
5
6  // Calculate chi-square
7  x2 = ((n-1)*s.sd^2)/(sigma^2)
8
9  // Calculate critical value
10 x2.alpha = qchisq(alpha, df=n-1)
11
12
13
14
15
16
17
18
19
20
21
22
```

```
> x2
[1] 5.259313
> x2.alpha
[1] 2.732637
```

Hypothesis Testing 1 Sample

Population Variance: Two-Tailed

WHAT WE KNOW?

- Sample size, n : 9
- Sample standard deviation: 3
- $H_0: \sigma = 3.7$ $H_1: \sigma < 3.7$
- Significance level: 0.05

qchisq() is used to get calculate chi-square statistics

2 parameters: 1) significance level, 2) df

Ref: <http://personal.psu.edu/drh20/astrostatistics/excerpts/html/stats/html/Chisquare.html>

```

1  n = 9
2  s.sd = 3
3  sigma = 3.7
4  alpha = 0.05
5
6  // Calculate chi-square
7  x2 = ((n-1)*s.sd^2)/(sigma^2)
8
9  // Calculate critical value
10 x2.alpha = qchisq(alpha, df=n-1)
11
12
13
14
15
16 > x2
17 [1] 5.259313
18
19 > x2.alpha
20 [1] 2.732637
21
22
  
```



References

- <http://www.r-tutor.com/>

THE END...